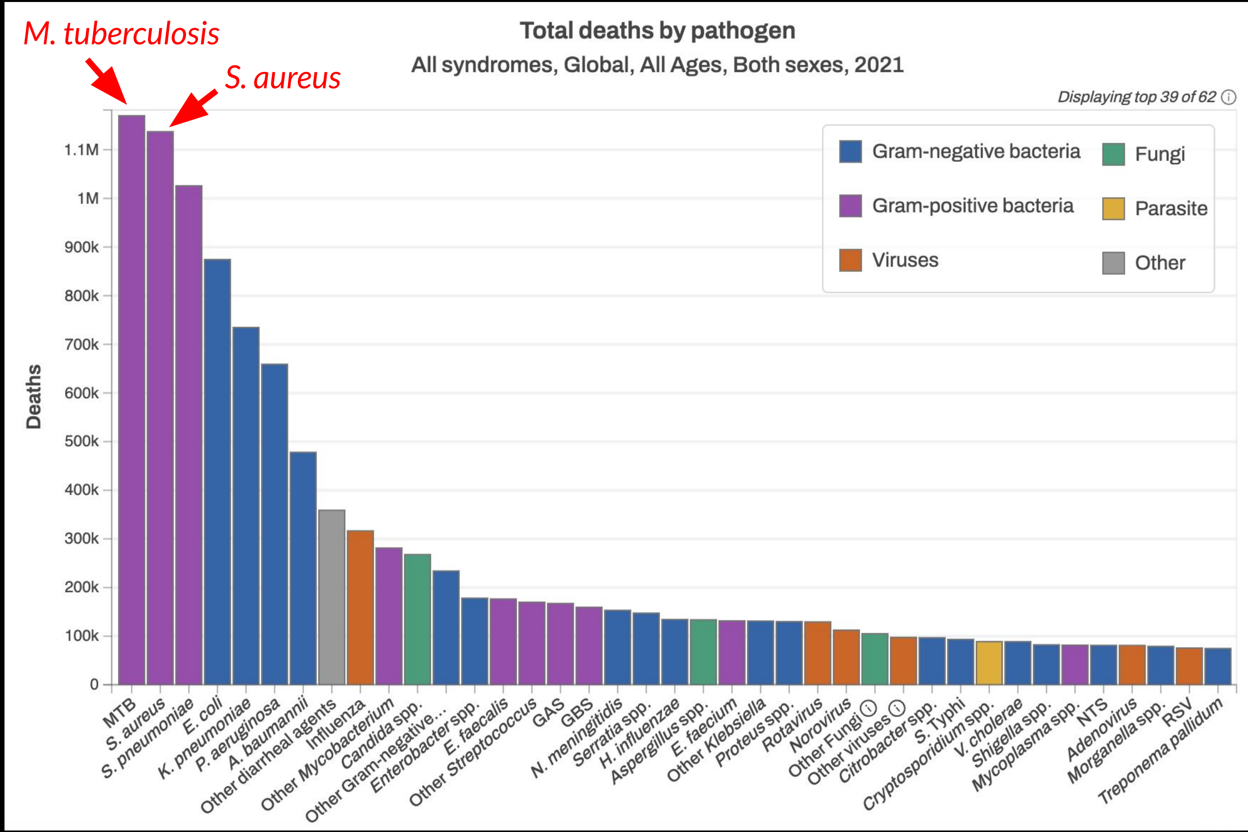
Multi-omics data and sample-efficient model architectures improve prediction of bacterial T cell antigens

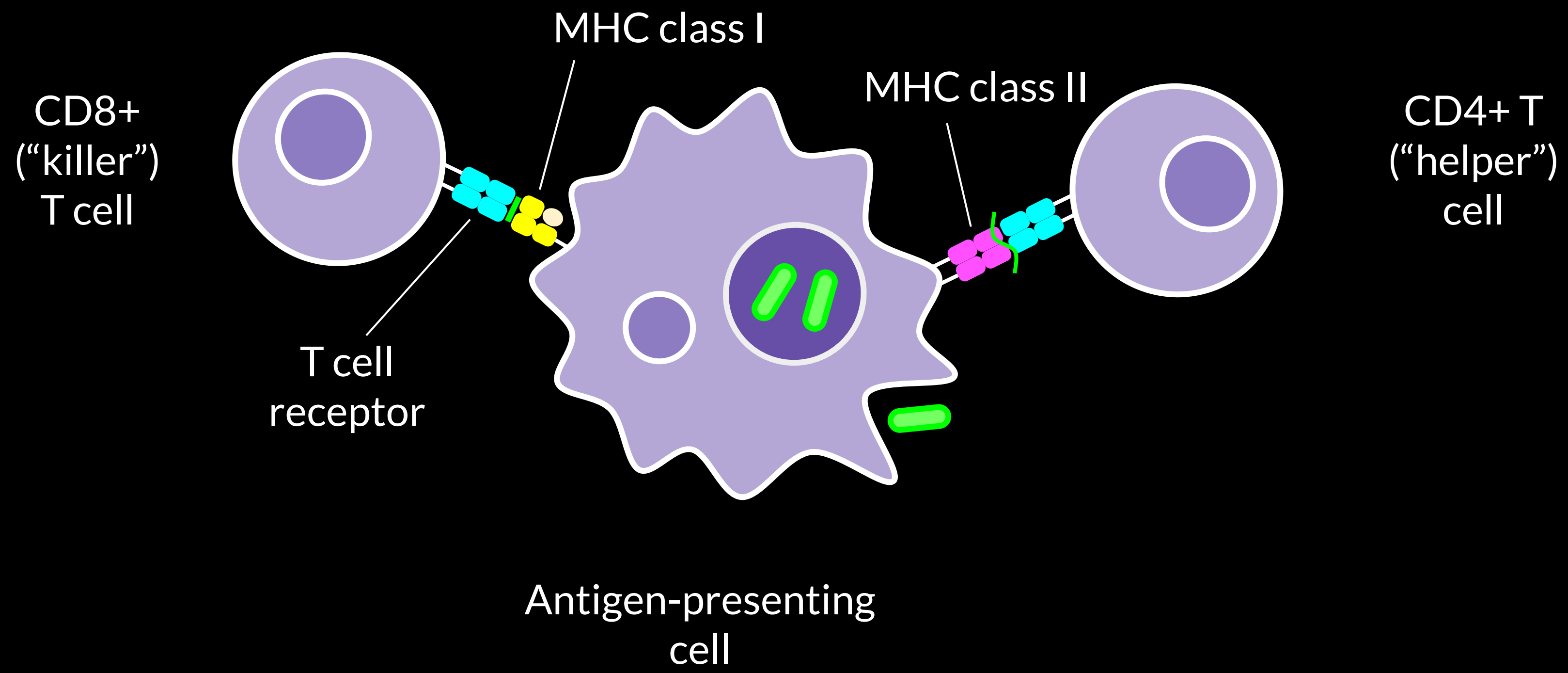
Owen Leddy
UCSD Department of Pediatrics
Laboratory of George Liu

Effective vaccines are urgently needed for bacterial pathogens



Naghavi et al. *The Lancet* 2024

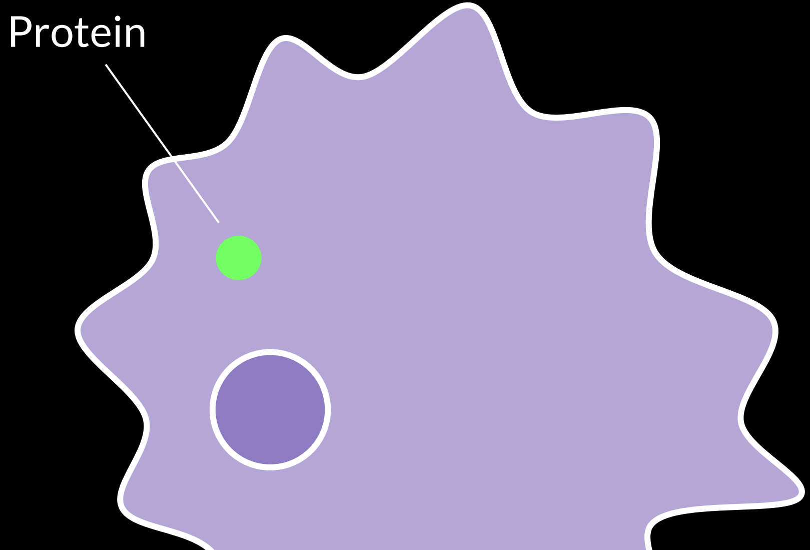
T cells recognize peptide antigens presented on MHCs to mediate immunity to bacterial infection



Cells present fragments of internal proteins on the cell surface on MHCs for immune surveillance

Protein sequence

MTEQQWNFAGIEAAASAIQGNVTSIHSLLDEGKQSLTKLAAA
WGGSGSEAYQGVQQKWDATATELNNALQNLARTISEAGQAMA
STEGNVTGMFA



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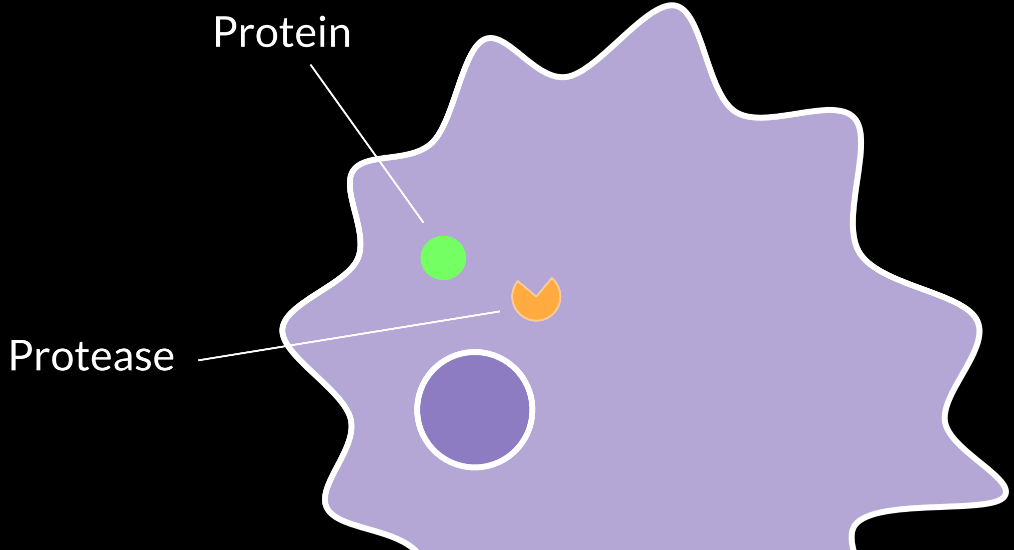

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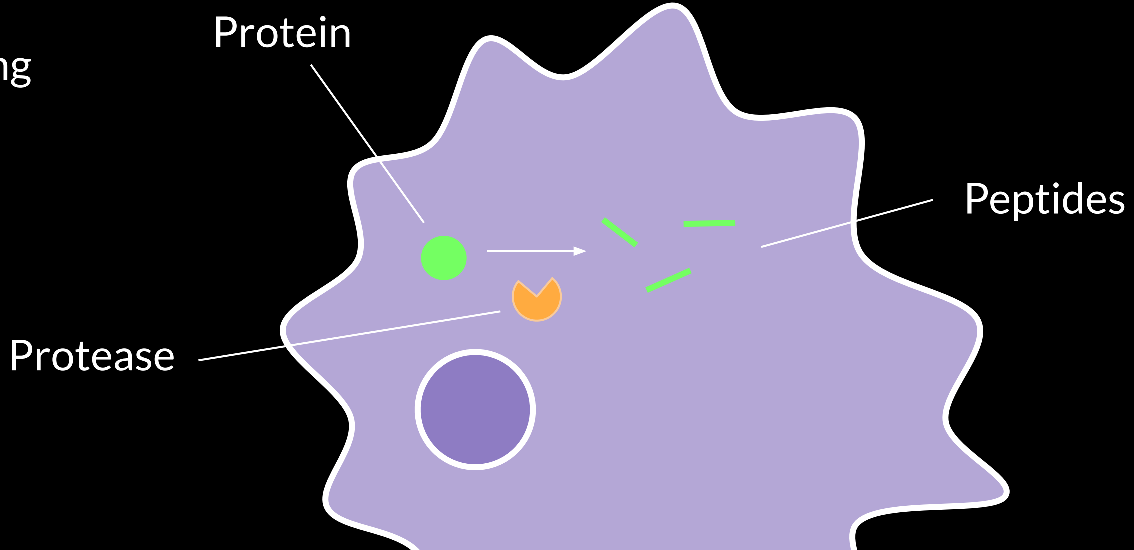
↓ Protease processing

Peptide sequence

LLDEGKQSL



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STEGNVTGMFA

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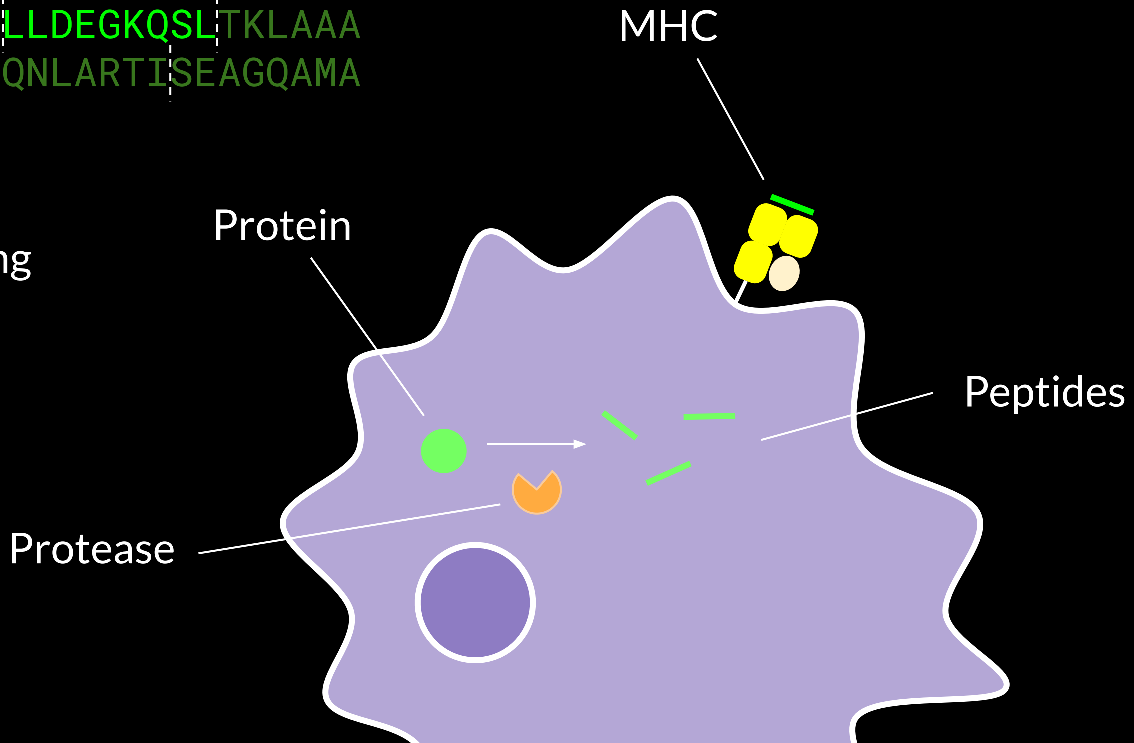
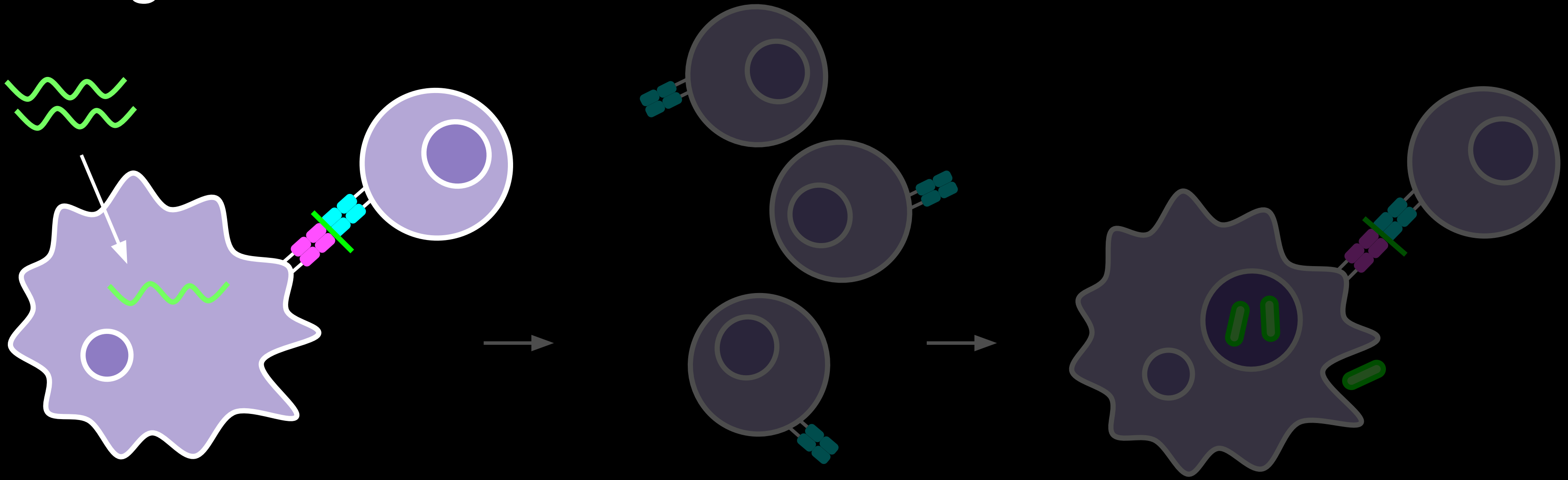


Vaccines can prime T cells to recognize a target protein antigen



Macrophages and
dendritic cells present
vaccine antigen to T cells

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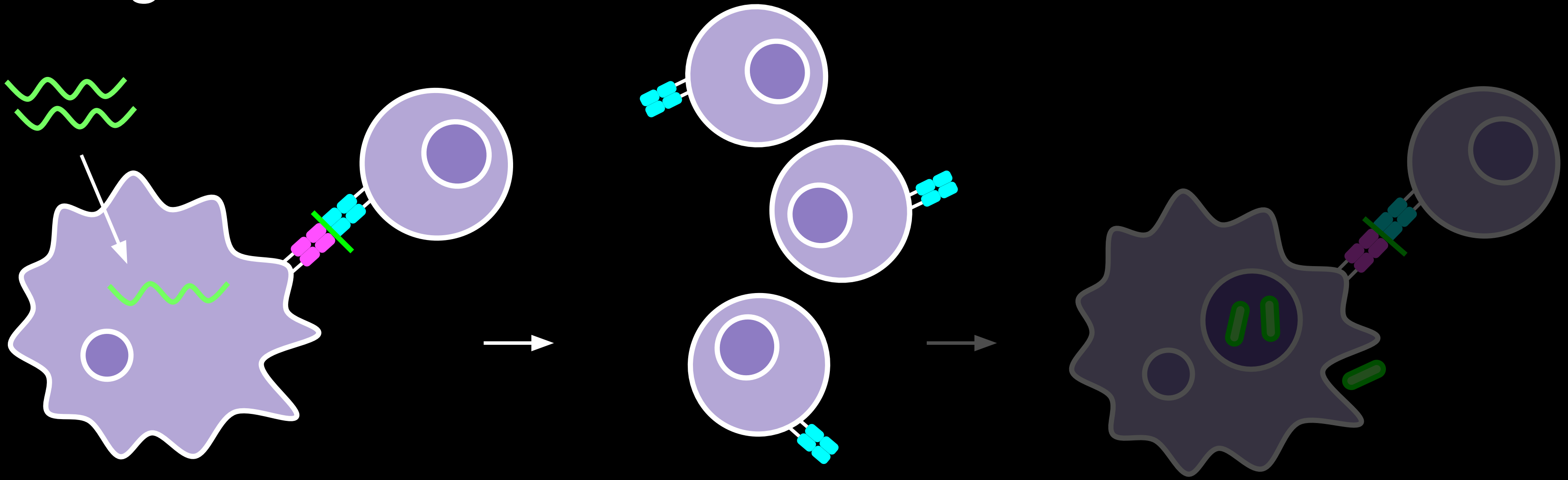
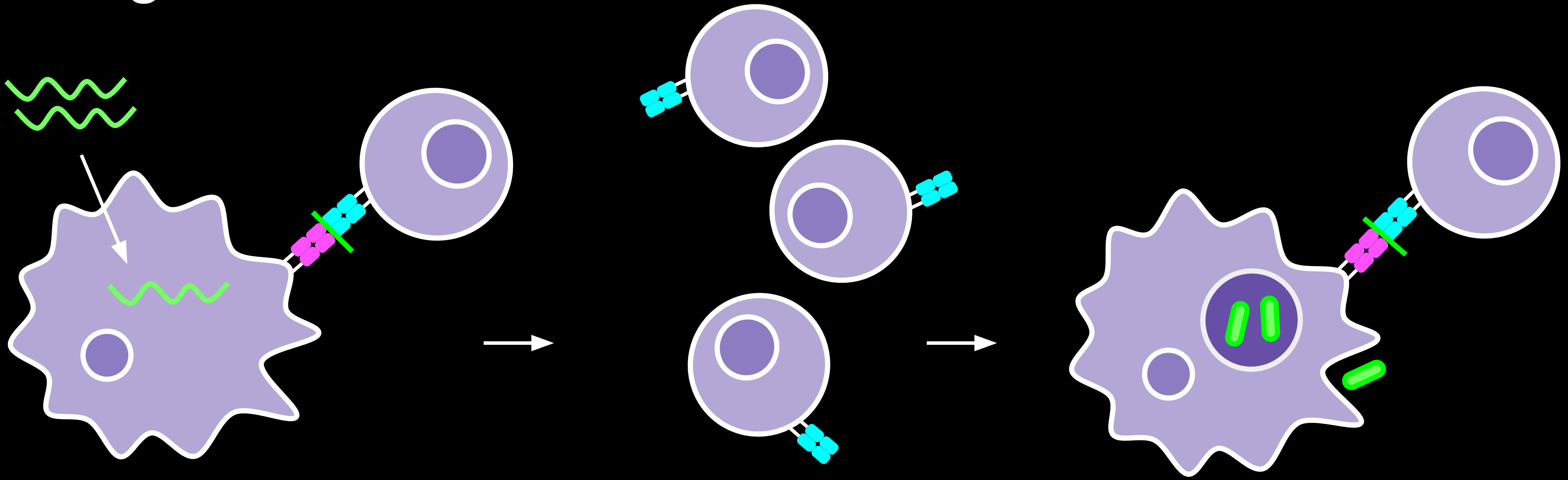


Macrophages and dendritic cells present vaccine antigen to T cells

T cells activate and proliferate

Vaccines can prime T cells to recognize a target protein antigen



Macrophages and dendritic cells present vaccine antigen to T cells

T cells activate and proliferate

T cells are ready to recognize infected cells if an infection occurs

How do we determine which antigens to put in a vaccine for TB or MRSA?

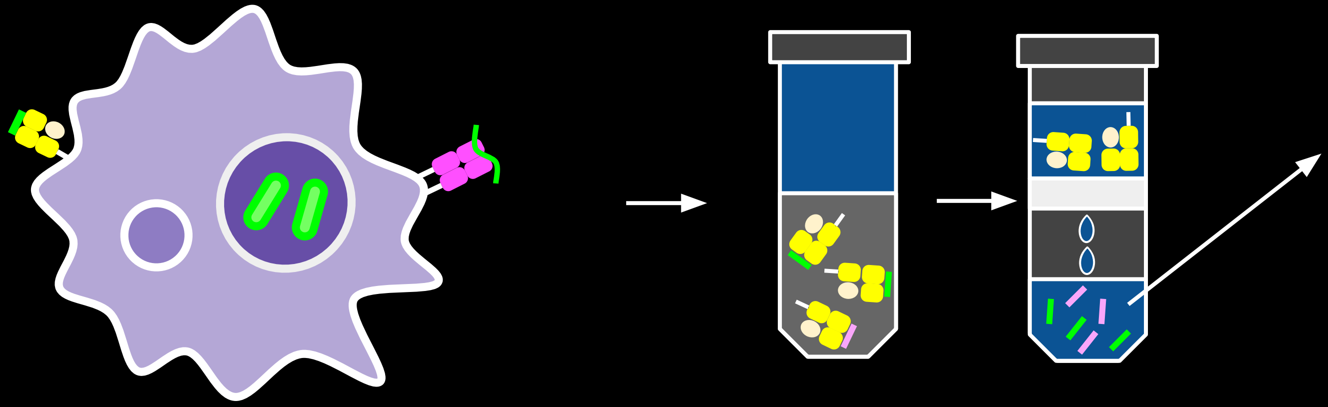
SARS-CoV-2
(cause of COVID-19)
29 genes

How do we determine which antigens to put in a vaccine for TB or MRSA?

SARS-CoV-2
(cause of COVID-19)
29 genes

M. tuberculosis:
~4000 genes

Mass spectrometry (MS) enables identification of peptides presented on MHCs



KLGSVPVTV
DEAEVHLKF
TEFGPGTAY
LLDEGKQSL
QLIKELPEL
TPSQPLPTY
IEEMLQTAY
RIIDTVAQV
...

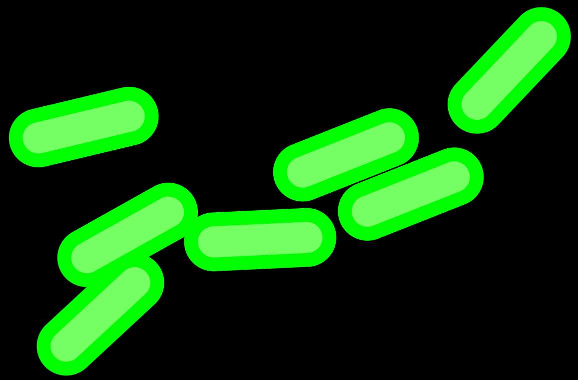
Antigen-presenting cell

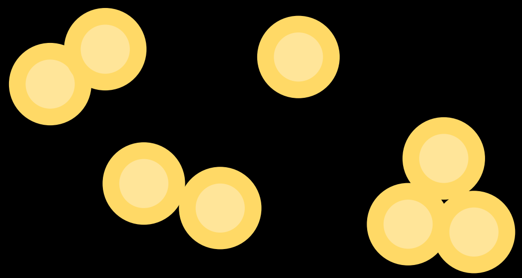
Isolate MHCs

Purify peptides

Mass spectrometry to sequence **host-** and **pathogen**-derived peptides

Overview

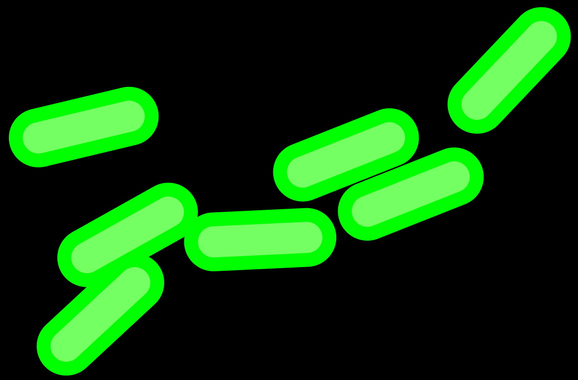


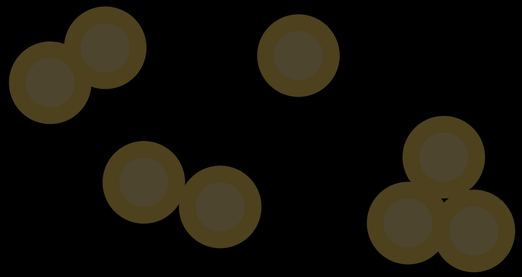


Mycobacterium tuberculosis (Mtb)

Staphylococcus aureus (SA)

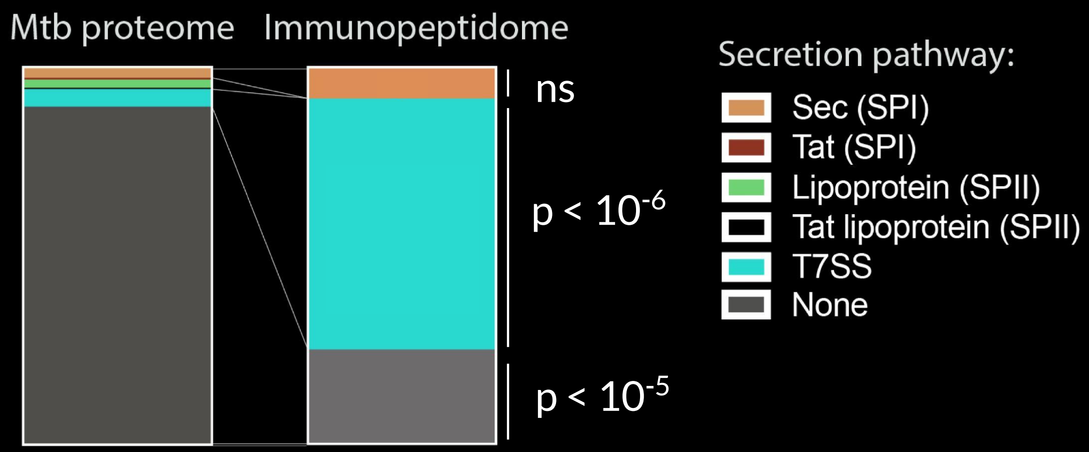
Overview

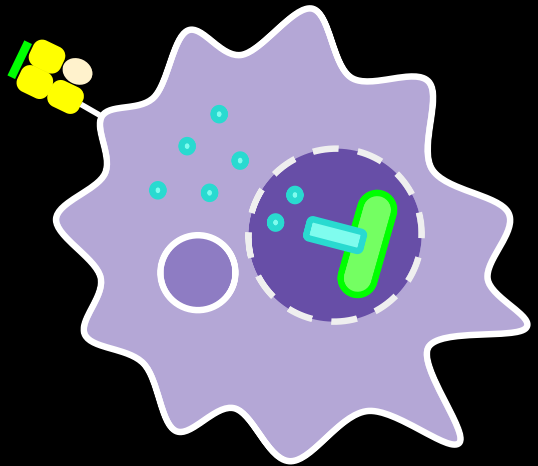




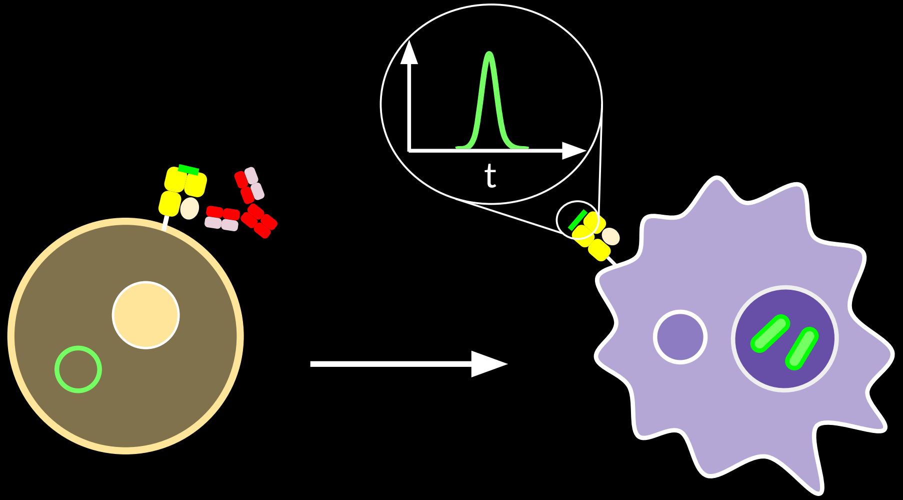
Mycobacterium tuberculosis (Mtb)

Mtb peptides presented on MHC-I derive from proteins exported by type VII secretion systems (T7SS)





MHC binding assays and protein localization are not sufficient to predict peptide presentation

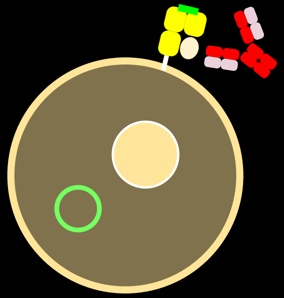


High-throughput MHC
binding assays
(yeast display)

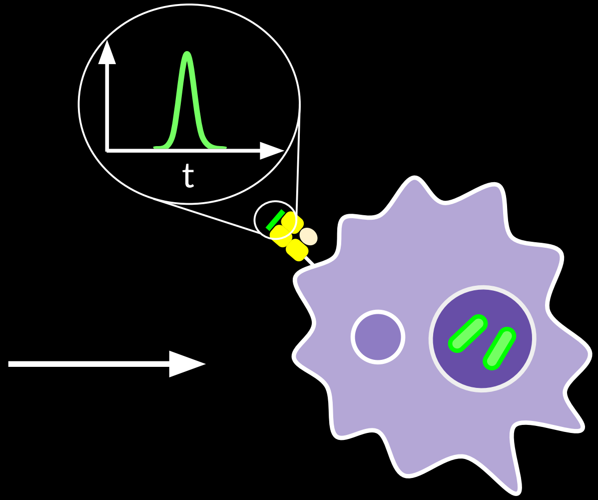
Targeted MS validation

With Patrick Holec and Michael Birnbaum

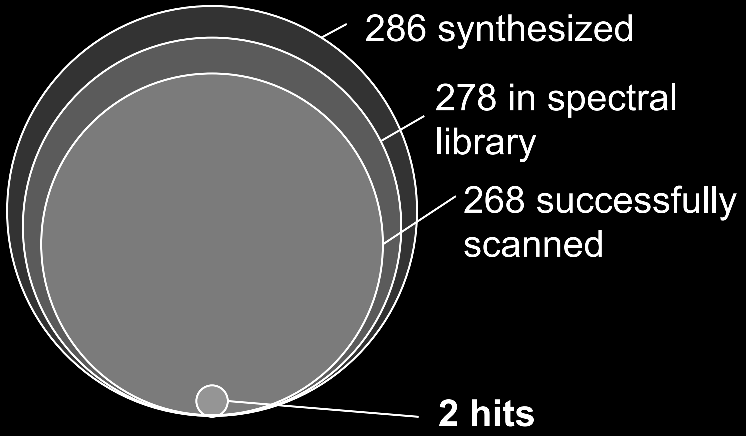
MHC binding assays and protein localization are not sufficient to predict peptide presentation



High-throughput MHC binding assays (yeast display)

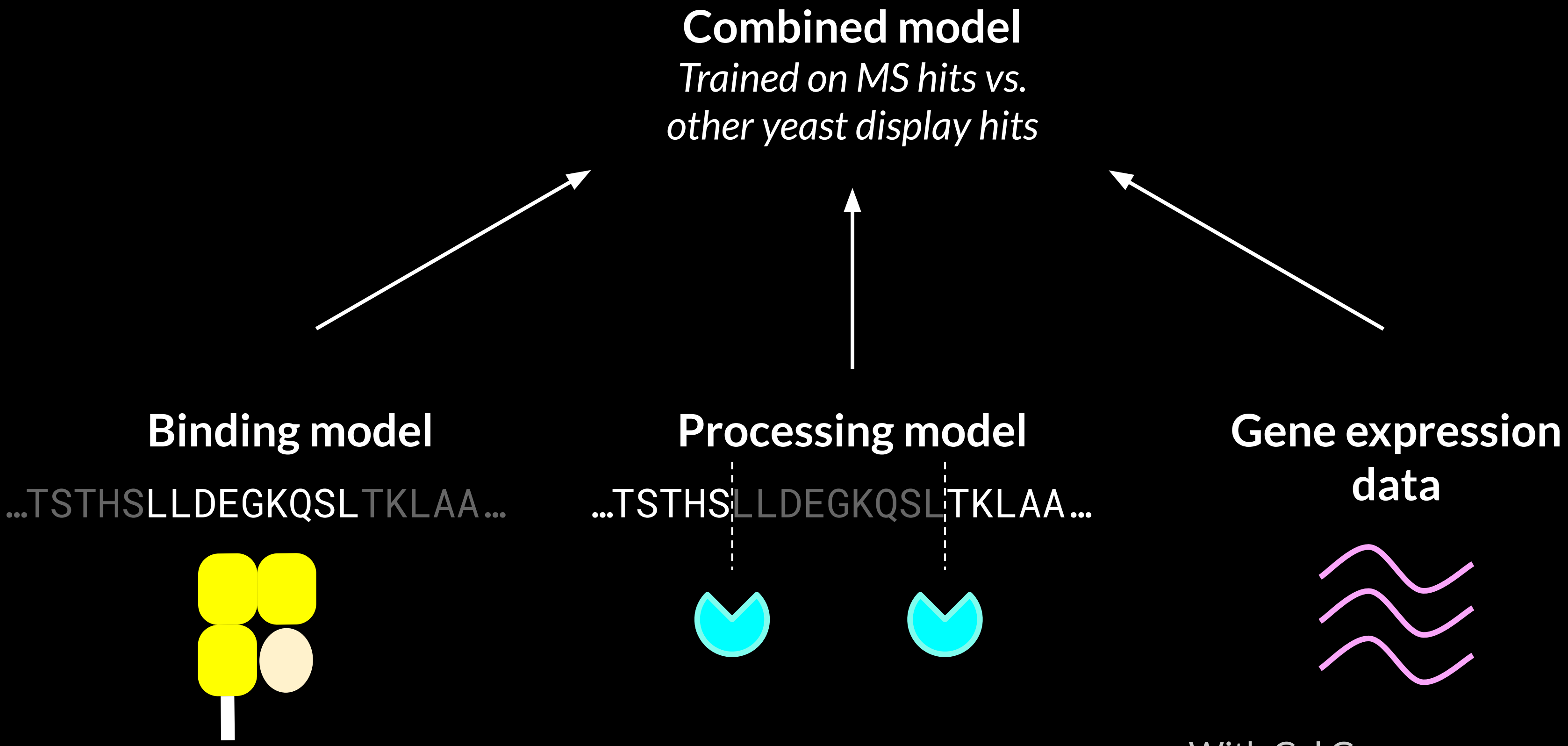


Targeted MS validation



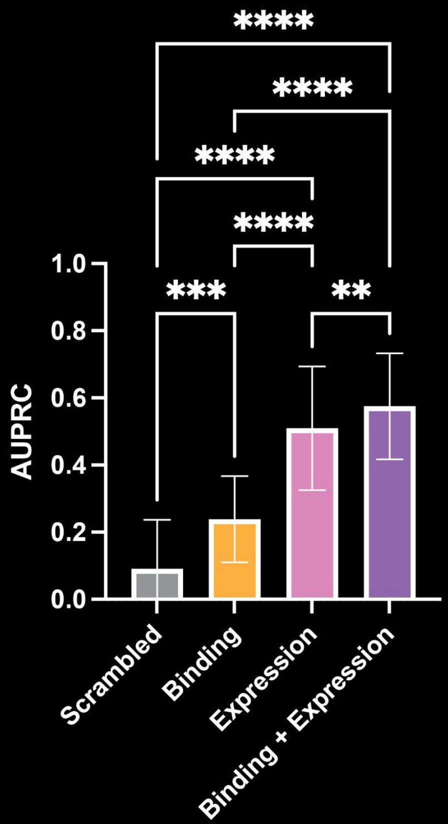
With Patrick Holec and Michael Birnbaum

Multiple factors may influence *Mtb* antigen presentation

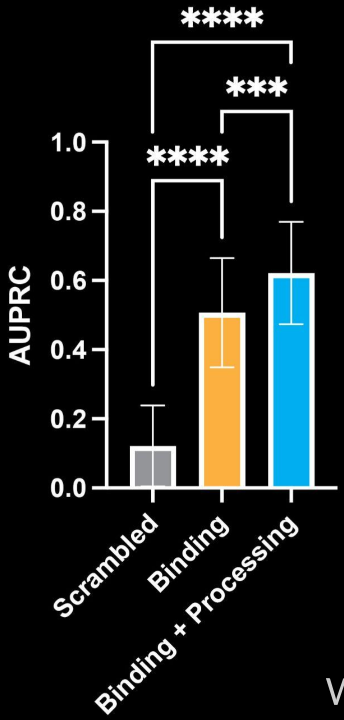


With Cal Gunnarsson and David Ren

Gene expression level and proteolytic processing influence MHC peptide presentation



High-expression antigens



With Cal Gunnarsson and David Ren

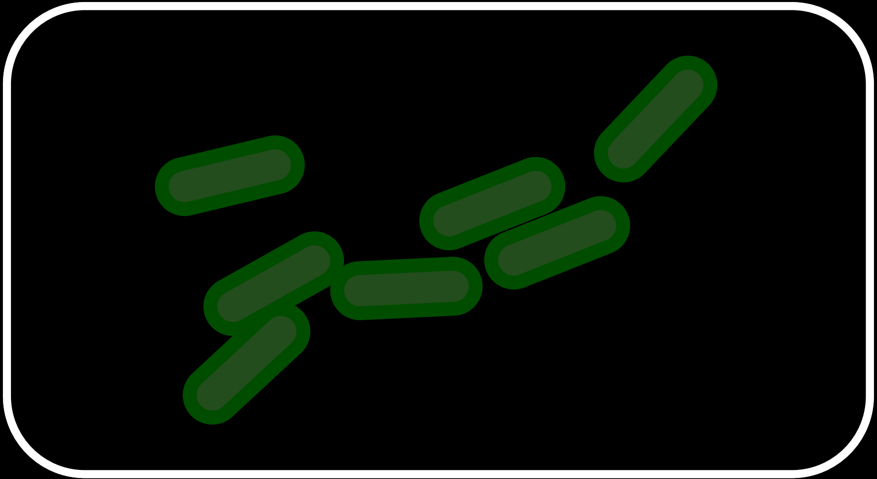
Experimental validation of novel *Mtb* MHC peptides

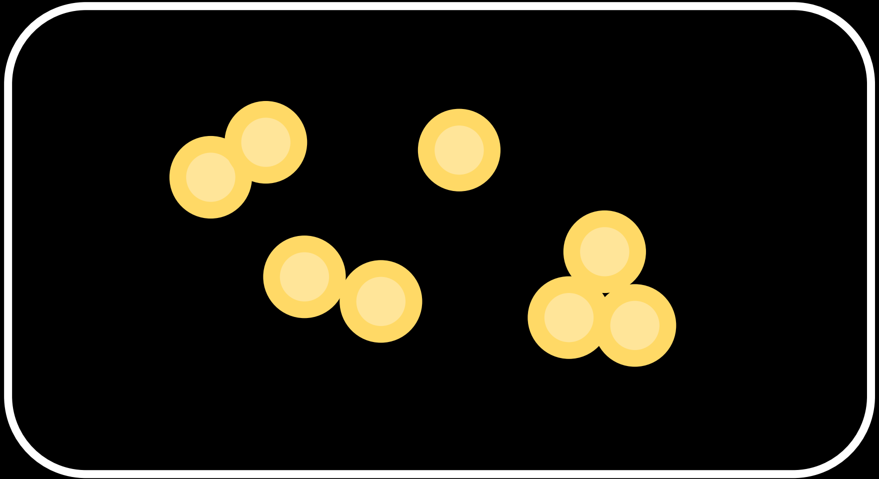
	Peptide sequence	Protein	Detected?
Positive control	VYLTAHNAL	EspC	☐
High-scoring	AYQGVQQKW	EsxA	☐
	AFQAAHARF	EsxG	☐
	RAAAIHDQF	PE13	☐
	DFTFLDAIF	PPE51	☐
	AYGSVLSGL	PPE51	☐
	GLSQVTGLF	EspA	☐
Low-scoring	QFLEETFAA	PPE46	☐
	RFPGFYIPF	PE4	☐
	LVTALFGAP	PE_PGRS2	☐
	AYQQRFVLA	PE_PGRS45	☐

Experimental validation of novel *Mtb* MHC peptides

	Peptide sequence	Protein	Detected?
Positive control (1/1)	VYLTAHNAL	EspC	
High-scoring (2/6)	AYQGVQQKW	EsxA	
	AFQAAHARF	EsxG	
	RAAAIHDQF	PE13	
	DFTFLDAIF	PPE51	
	AYGSVLSGL	PPE51	
	GLSQVTGLF	EspA	
Low-scoring (0/4)	QFLEETFAA	PPE46	
	RFPGFYIPF	PE4	
	LVTALFGAP	PE_PGRS2	
	AYQQRFVLA	PE_PGRS45	

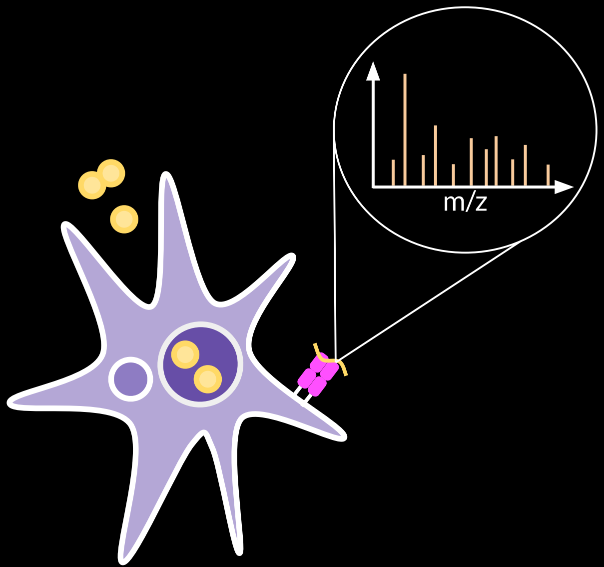
Overview

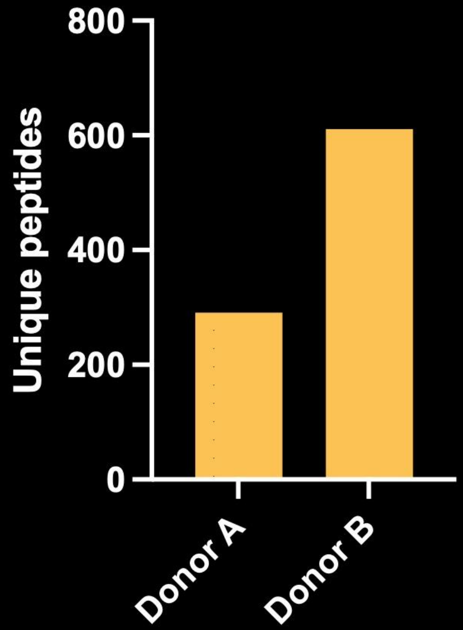




Staphylococcus aureus (SA)

Identification of *S. aureus* MHC-II peptides

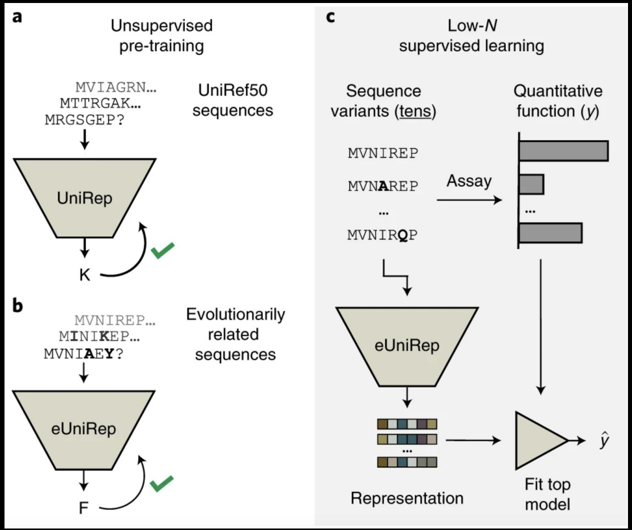




With Jacquelyn Castañeda, Igor Wierzbicki, Irfan Baig, David Gonzalez, and Majid Ghassemian

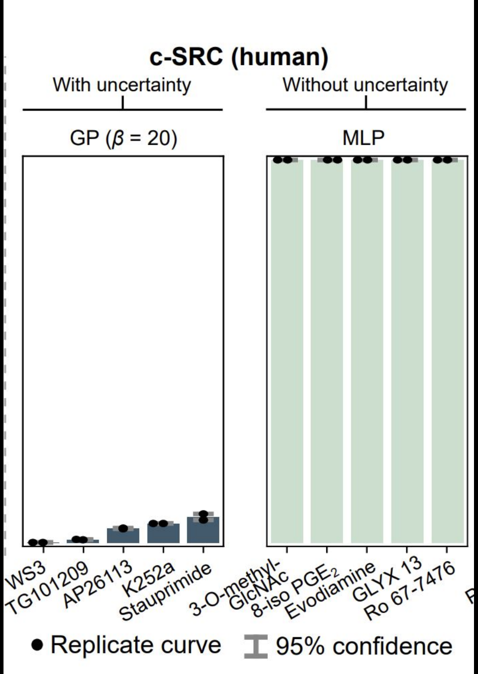
If we need pathogen-specific ML models, how do we make them more data-efficient?

Transfer learning



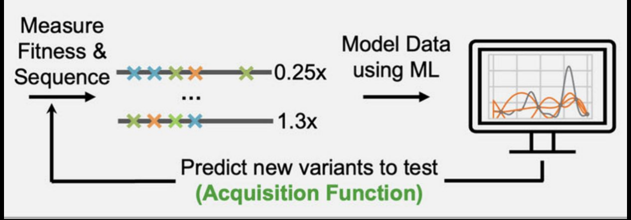
Biswas et al. *Nature Communications* 2021

Bayesian inference



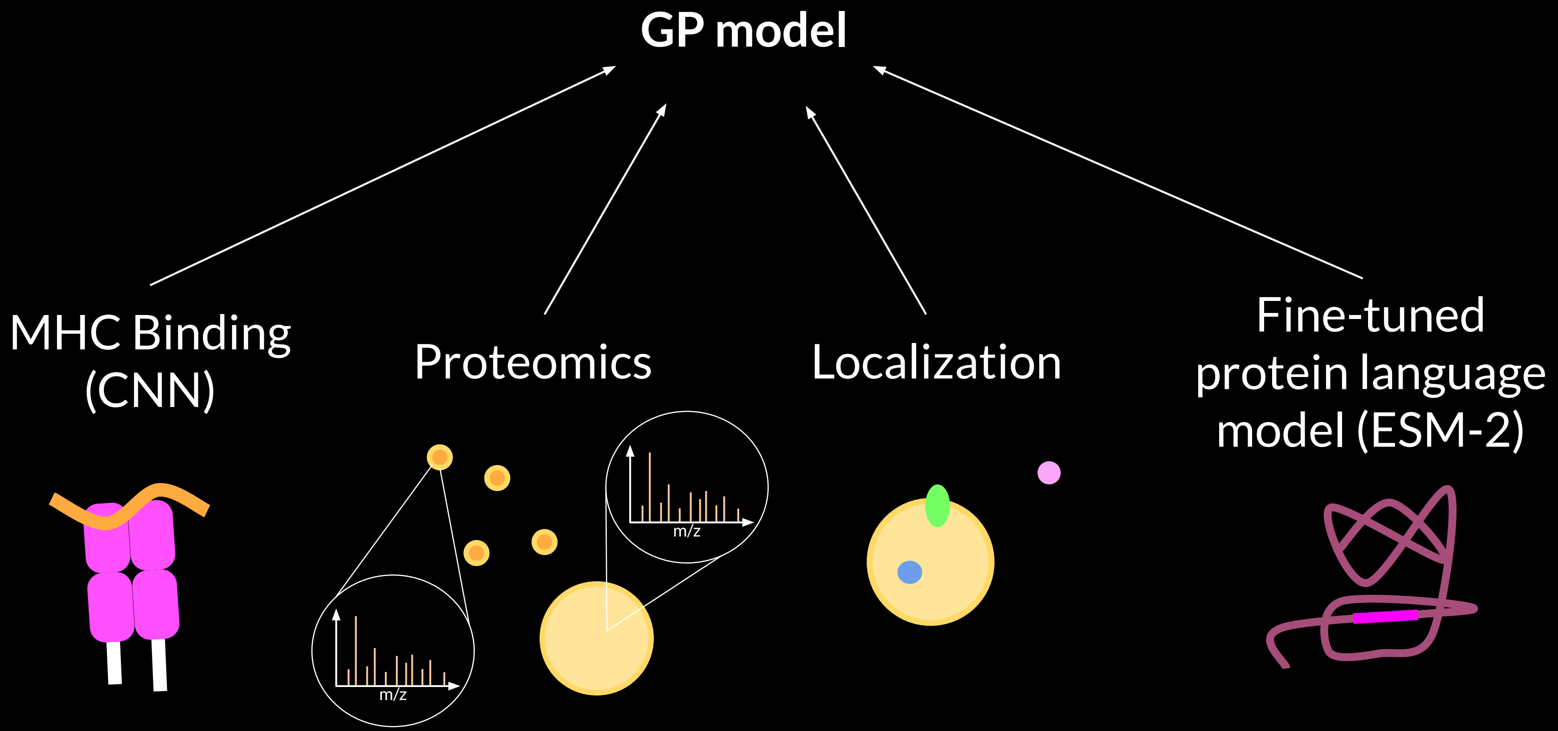
Hie et al. *Cell Systems* 2020

Active learning

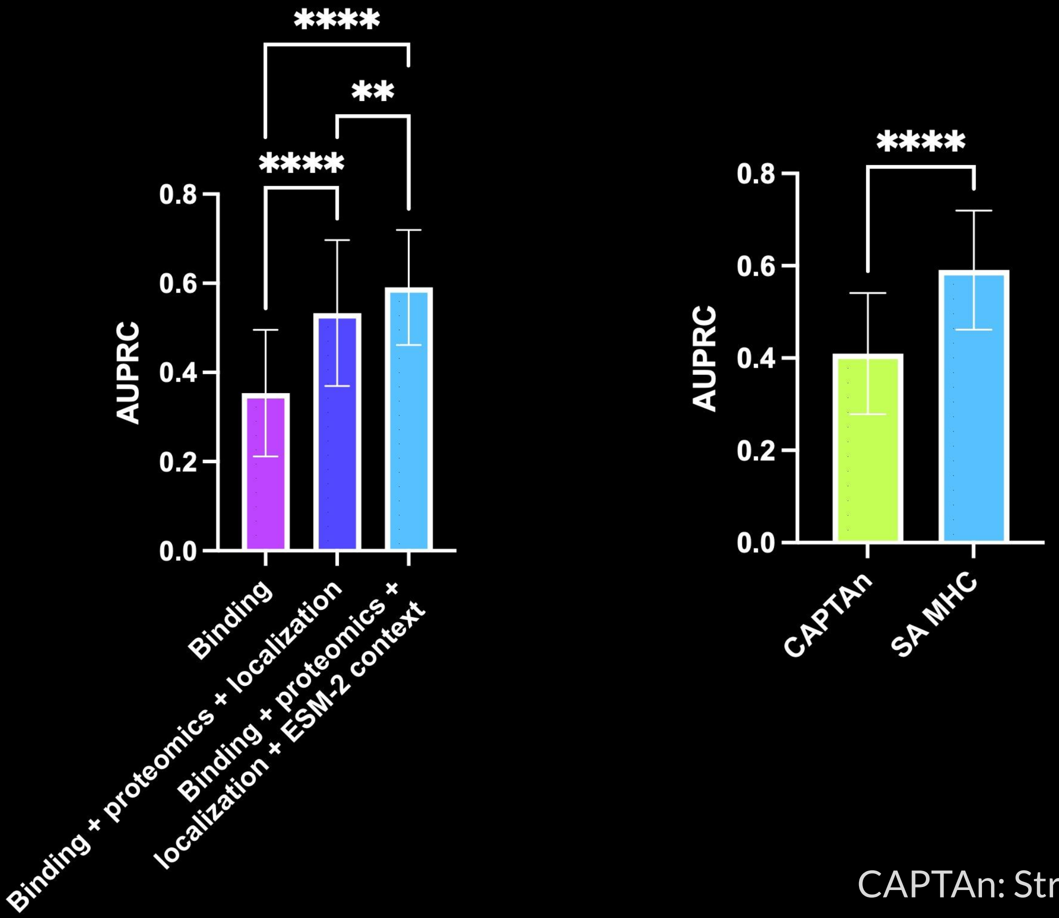


Yang et al. *Nature Communications* 2025

Gaussian process models to predict bacterial peptide presentation

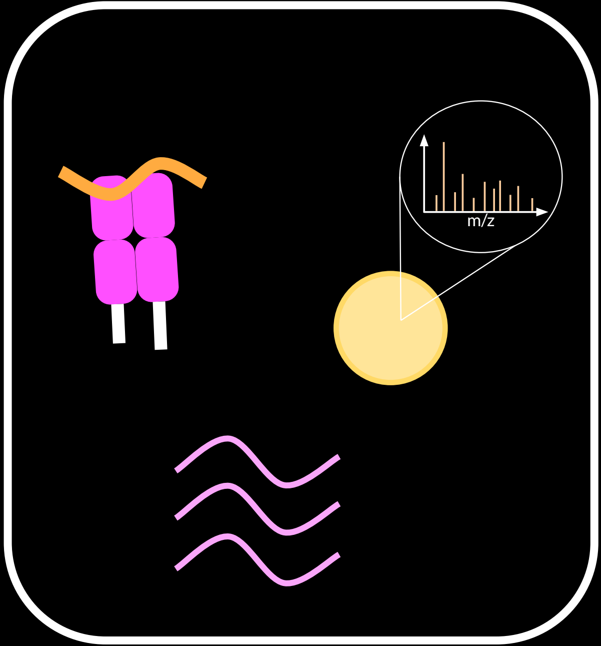


Proteomic data and transfer learning improve MHC peptide prediction

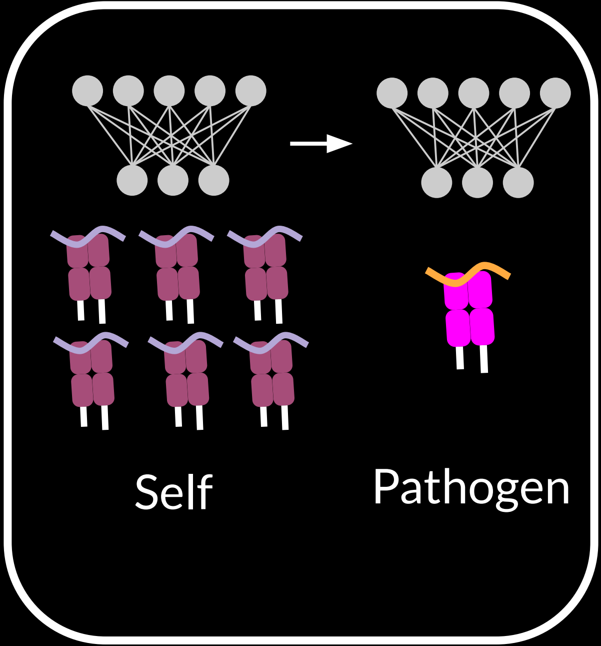


CAPTAn: Stražar et al. *Immunity* 2023

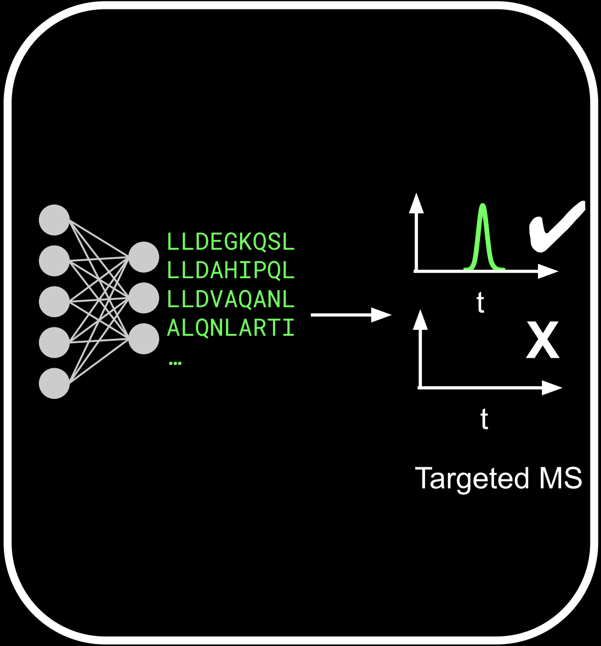
Conclusions



Combining multiple systems biology modalities helps predict MHC peptides

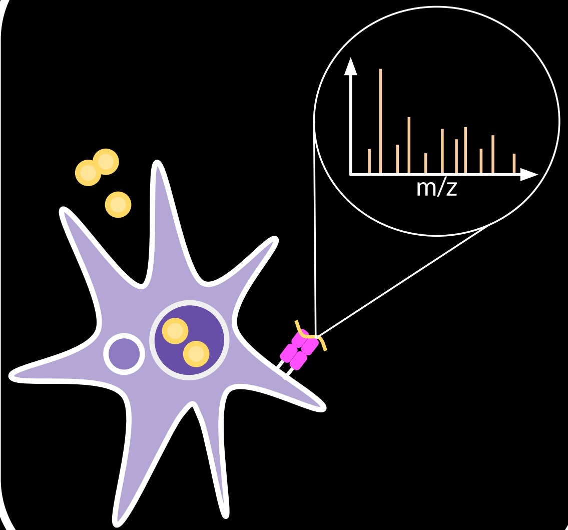


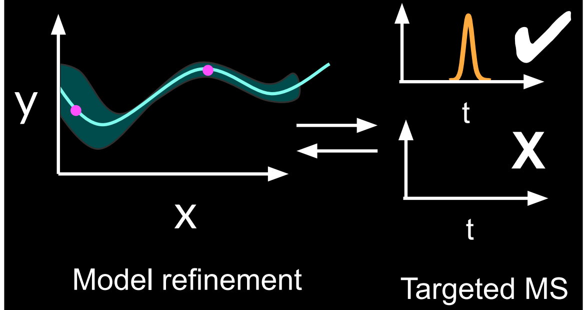
Pre-training on large datasets of self peptides facilitates learning pathogen-specific rules

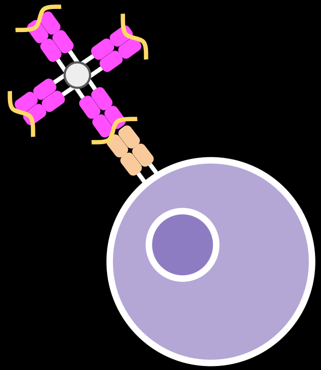


ML models can nominate MHC peptides for targeted MS validation

Future directions







Experimental validation of *S. aureus* MHC peptide predictions

Bayesian active learning

Analyze *in vivo* T cell response

Acknowledgements

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NIH
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NIH
NIAID

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MIT BE
BIOLOGICAL ENGINEERING

Ragon Institute
MASS GENERAL | MIT | HARVARD

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